Assignment 10: Data Scraping

Liam Connolly

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Load the packages `tidyverse`, `rvest`, and any others you end up using.
- Check your working directory

```
#1
library(tidyverse)
library(rvest)
setwd("/home/guest/EDE_Fall2025/Data/Assignment10")
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025/Data/Assignment10"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham's 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to https://www.ncwater.org/WUDC/app/LWSP/search.php
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```r
#2
url <- "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"
page <- read_html(url)
```

3. The data we want to collect are listed below:

- From the "1. System Information" section:
- Water system name
- PWSID
- Ownership
- From the "3. Water Supply Sources" section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

> HINT: The first value should be "Durham", the second "03-32-010", the third "Municipality", and the last should be a vector of 12 numeric values (represented as strings)".

```r
#3
system_info <- page %>%
  html_elements("table") %>%
  .[[1]] %>%
  html_table(fill = TRUE)
water_system_name <- system_info[1,2]
pwsid             <- system_info[2,2]
ownership         <- system_info[3,2]
tables <- page %>% html_elements("table")
max_day_table <- tables[[13]] %>% html_table(fill = TRUE)
print(names(max_day_table))
```

```
## [1] ""                         "Average DailyUse (MGD)" "Max DayUse (MGD)"
## [4] ""                         "Average DailyUse (MGD)" "Max DayUse (MGD)"
## [7] ""                         "Average DailyUse (MGD)" "Max DayUse (MGD)"
```

```r
print(head(max_day_table))
```

```
## # A tibble: 4 x 9
##   ``    `Average DailyUse (MGD)` `Max DayUse (MGD)` ``    Average DailyUse (MG~1
##   <chr>                    <dbl>              <dbl> <chr>                  <dbl>
## 1 Jan                       29.0               34.5 May                     31.6
## 2 Feb                       29.4               32.1 Jun                     37.7
## 3 Mar                       32.5               38.2 Jul                     35.9
## 4 Apr                       31.4               36.7 Aug                     37.5
## # i abbreviated name: 1: `Average DailyUse (MGD)`
## # i 4 more variables: `Max DayUse (MGD)` <dbl>, `` <chr>,
## #   `Average DailyUse (MGD)` <dbl>, `Max DayUse (MGD)` <dbl>
```

```r
max_day_use <- max_day_table$`Max DayUse (MGD)`
str(max_day_use)
```

```
##  num [1:4] 34.5 32.1 38.2 36.7
```

```r
max_day_use
```

```
## [1] 34.50 32.10 38.20 36.73
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

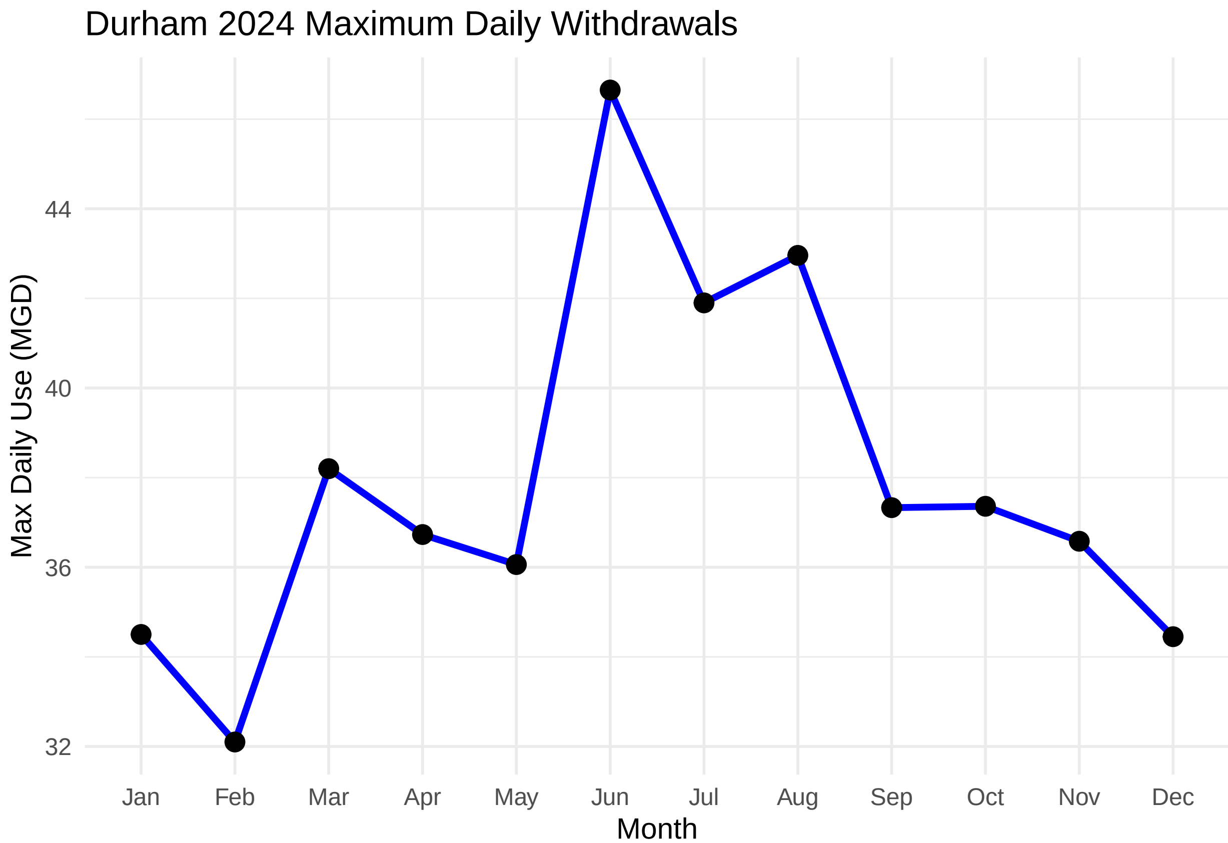
   TIP: Use `rep()` to repeat a value when creating a dataframe.

   NOTE: It's likely you won't be able to scrape the monthly widthrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc…

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```r
#4
tables <- page %>% html_elements("table") %>% map(html_table, fill = TRUE)
four_row_tables <- keep(tables, ~ nrow(.x) == 4)
max_day_table <- four_row_tables[[which(sapply(four_row_tables, ncol) == 9)]]
max_long <- tibble(
  Month = c(max_day_table[[1]], max_day_table[[4]], max_day_table[[7]]),
  MaxDay_MGD = c(max_day_table[[3]], max_day_table[[6]], max_day_table[[9]])
)
df <- tibble(
  Water_System = rep(water_system_name, 12),
  PWSID        = rep(pwsid, 12),
  Ownership    = rep(ownership, 12),
  Month        = max_long$Month,
  MaxDay_MGD   = max_long$MaxDay_MGD,
  Year         = 2024
) %>%
  mutate(
    # Converts abbreviated month names to proper dates
    Date = as.Date(paste(Month, Year, "1"), format = "%b %Y %d")
  )
#5
df %>%
  mutate(Month = factor(Month, levels = month.abb)) %>%
  ggplot(aes(x = Month, y = MaxDay_MGD, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(size = 3) +
  labs(
    title = "Durham 2024 Maximum Daily Withdrawals",
```

```
    x = "Month",
    y = "Max Daily Use (MGD)"
  ) +
  theme_minimal()
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - "PWSID" and "year" - that:

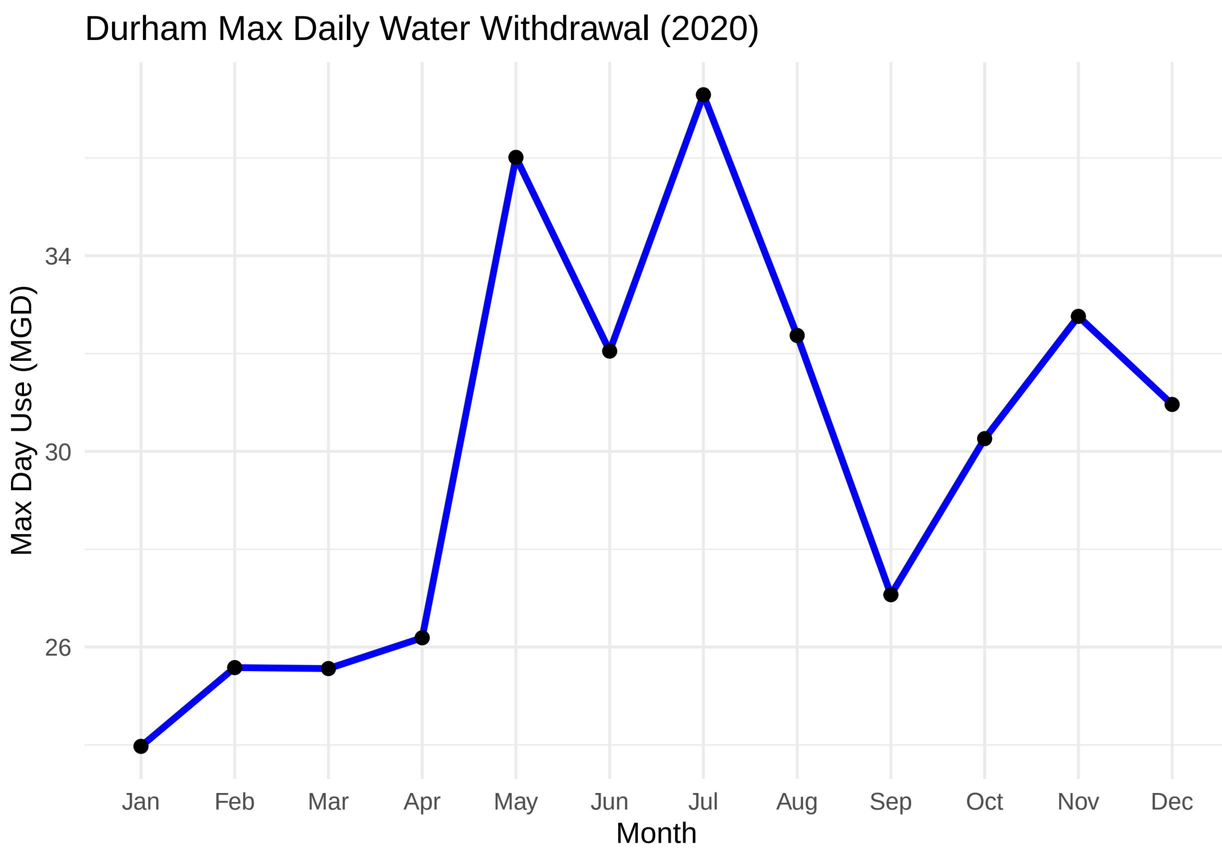
- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```
#6.
scrape_LWSP <- function(PWSID, year) {
  url <- paste0("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=",
                PWSID, "&year=", year)
  page <- read_html(url)
  system_name <- "Durham"
  ownership   <- "Municipality"
  tables <- page %>% html_elements("table") %>% map(html_table, fill = TRUE)
  max_day_table <- keep(tables, ~ any(grepl("Max Day", names(.x),
                                              ignore.case = TRUE)))[[1]]
```

```r
  text_block <- max_day_table %>% as.character() %>% paste(collapse = " ")
  maxday <- stringr::str_extract_all(text_block,
                       "\\d+\\.?\\d*")[[1]] %>% as.numeric()
  maxday <- maxday[1:12]
  months <- month.abb
  tibble(
    Water_System = rep(system_name, 12),
    PWSID = rep(PWSID, 12),
    Ownership = rep(ownership, 12),
    Month = factor(months, levels = month.abb),
    MaxDay_MGD = maxday,
    Year = rep(year, 12),
    Date = as.Date(paste(year, months, "1"), format = "%Y %b %d")
  )
}
#I am completely befuddled by this, maybe this works???
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```r
#7
durham2020 <- scrape_LWSP("03-32-010", 2020)
ggplot(durham2020, aes(x = Month, y = MaxDay_MGD, group = 1)) +
  geom_line(linewidth = 1.2, color = "blue") +
  geom_point(size = 2) +
  theme_minimal() +
  labs(title = "Durham Max Daily Water Withdrawal (2020)",
       x = "Month", y = "Max Day Use (MGD)")
```



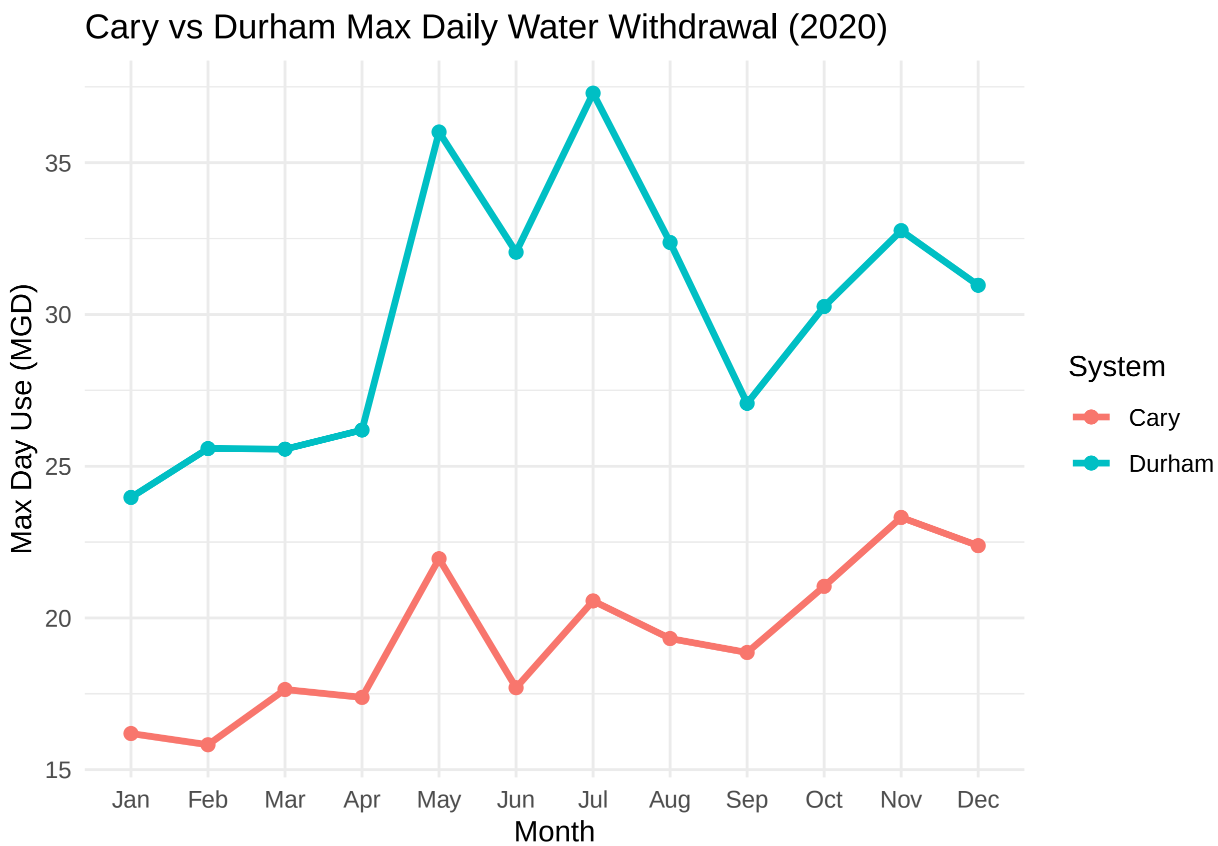

```r
glimpse(durham2020)
```

```
## Rows: 12
## Columns: 7
## $ Water_System <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "Durham~
## $ PWSID        <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-3~
## $ Ownership    <chr> "Municipality", "Municipality", "Municipality", "Municipa~
## $ Month        <fct> Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec
## $ MaxDay_MGD   <dbl> 23.97, 25.58, 25.56, 26.19, 36.01, 32.05, 37.29, 32.37, 2~
## $ Year         <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 202~
## $ Date         <date> 2020-01-01, 2020-02-01, 2020-03-01, 2020-04-01, 2020-05-0~
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```r
#8
cary2020 <- scrape_LWSP("03-92-020", 2020)
both2020 <- bind_rows(
  durham2020 %>% mutate(System = "Durham"),
  cary2020 %>% mutate(System = "Cary")
)
ggplot(both2020, aes(x = Month, y = MaxDay_MGD, color = System,
                     group = System)) +
  geom_line(linewidth = 1.2) +
```

```r
  geom_point(size = 2) +
  theme_minimal() +
  labs(title = "Cary vs Durham Max Daily Water Withdrawal (2020)",
       x = "Month", y = "Max Day Use (MGD)")
```




```r
glimpse(both2020)
```

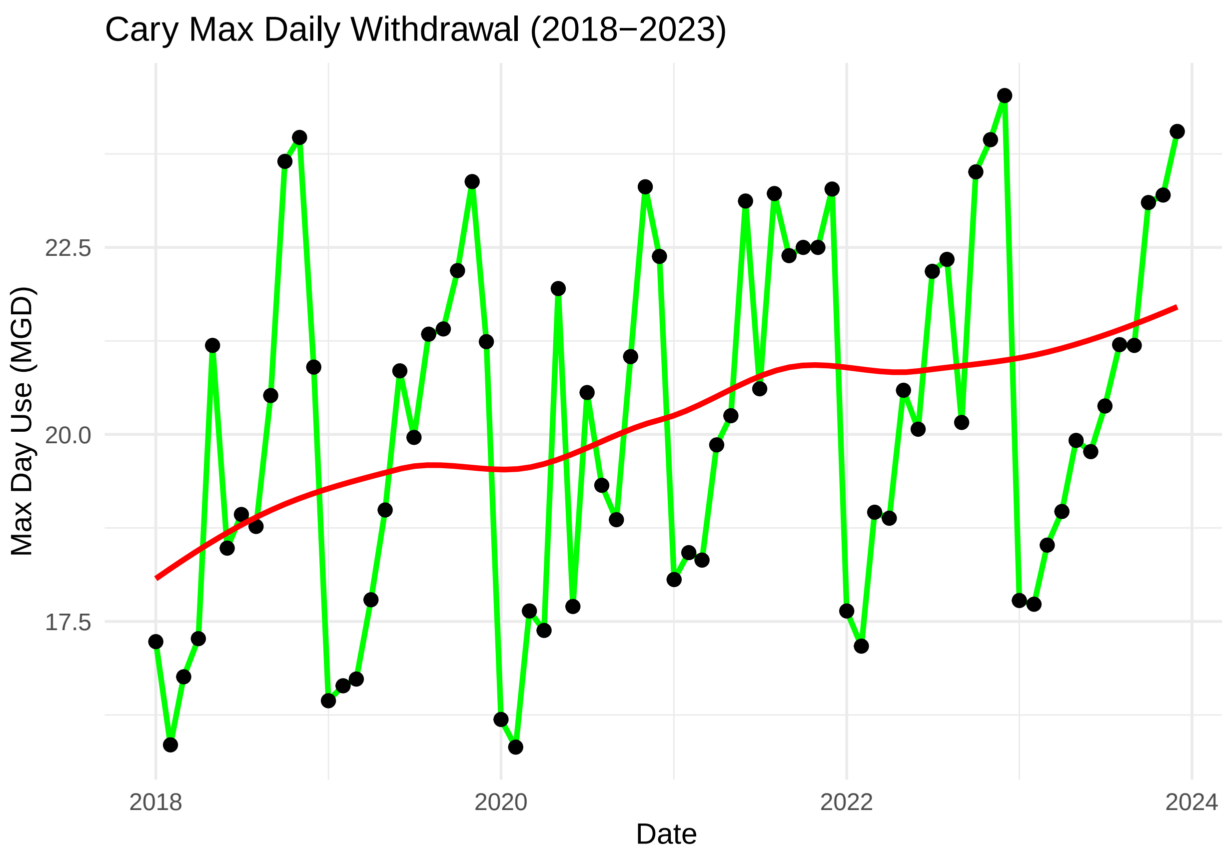
```
## Rows: 24
## Columns: 8
## $ Water_System <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "Durham~
## $ PWSID        <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-3~
## $ Ownership    <chr> "Municipality", "Municipality", "Municipality", "Municipa~
## $ Month        <fct> Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, De~
## $ MaxDay_MGD   <dbl> 23.97, 25.58, 25.56, 26.19, 36.01, 32.05, 37.29, 32.37, 2~
## $ Year         <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 202~
## $ Date         <date> 2020-01-01, 2020-02-01, 2020-03-01, 2020-04-01, 2020-05-~
## $ System       <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "Durham~
```

9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023.Add a smoothed line to the plot (method = 'loess').

   TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to `bind_rows()` to combine the dataframes into a single one, and use that to construct your plot.

```r
#9
years <- 2018:2023
cary_multi <- map(years, ~ scrape_LWSP("03-92-020", .x)) %>% bind_rows()
ggplot(cary_multi, aes(x = Date, y = MaxDay_MGD)) +
  geom_line(group = 1, linewidth = 1, color = "green") +
  geom_point(size = 2) +
  geom_smooth(method = "loess", color = "red", se = FALSE) +
  theme_minimal() +
  labs(title = "Cary Max Daily Withdrawal (2018-2023)",
       x = "Date", y = "Max Day Use (MGD)")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Cary's water usage shows a slight increase over the years. There are monthly fluctuations but, the pattern itself is that peak water demand is growing gradually. >